

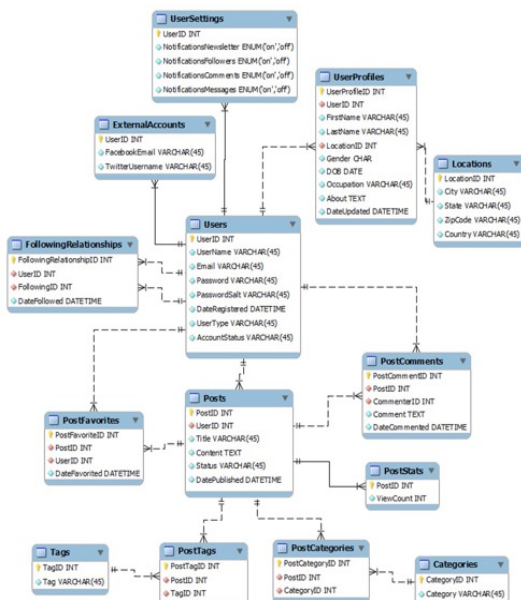
Training Graph Neural Networks on Top of Relational Databases

Self-Leverage Relational Data, Eliminating Highly Time-Consuming Dataset Preparation, Analysis, Feature Engineering, Target Labeling ..

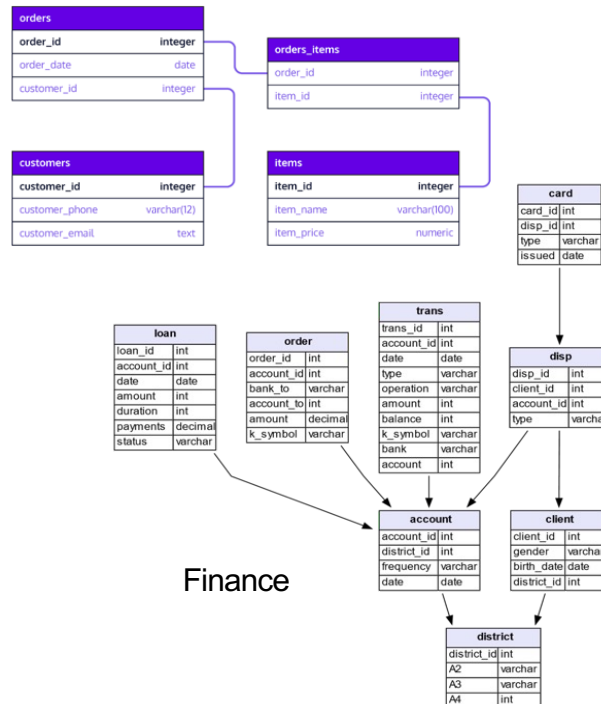
Prof. Gianluca Moro
Teaching Assistant: Dr. Giacomo Frisoni
DISI – University of Bologna
name.surname@unibo.it

Data stored in Relational Databases

Social Media



Commerce

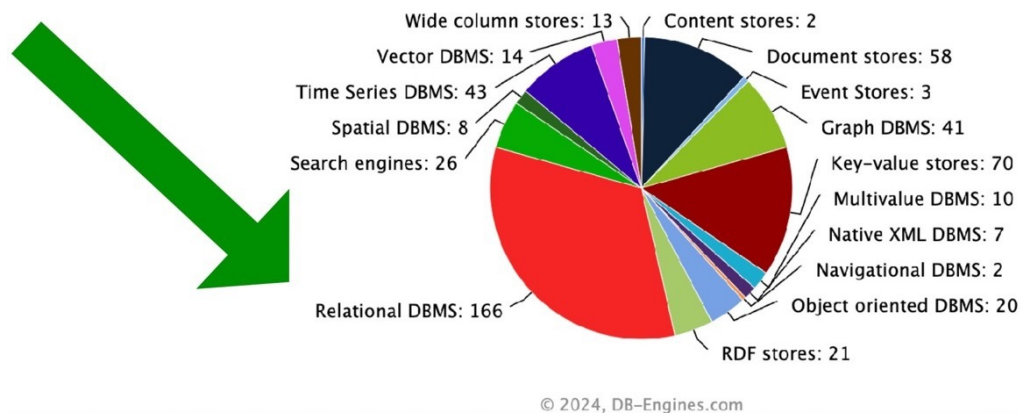


Finance

Database Management Systems

DBMS popularity broken down by database model

Number of systems per category, April 2024



Graph Neural Networks Meet Relational Databases

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Predictions on Relational Data

- Which **products** will a user **purchase** in the next 7 days?
- Will an **active user** **churn** in the next 90 days?
- What will be the **total sales** for each **product** in the next 30 days?

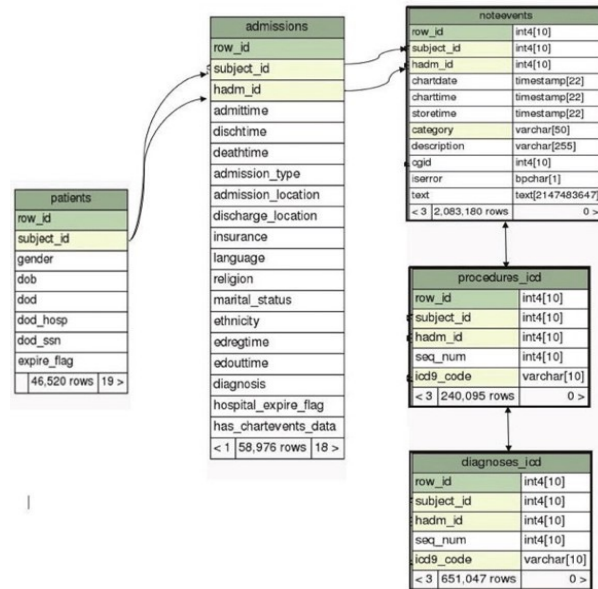


Graph Neural Networks Meet Relational Databases

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Predictions on Relational Data

- Will a **patient** return if discharged from the hospital?
- Which hospital admissions have **greatest risk to life** in the next 24 hours



Graph Neural Networks Meet Relational Databases

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Doing AI is slow & complex

Input data cleaning,
curation

Target Label
engineering

Feature
engineering

Architecture &
hyperparameter
search

ML Ops

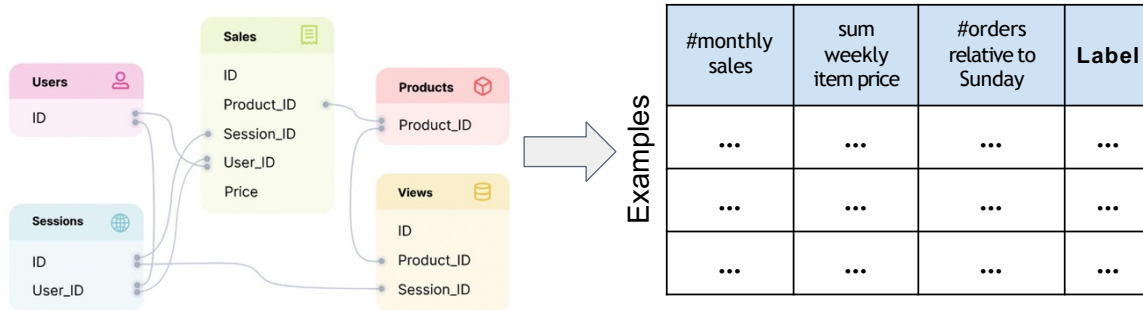
6-12 months of time for a team of data scientists,
data engineers and product engineers

Graph Neural Networks Meet Relational Databases

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Impedance Mismatch

Goal: Learn a **user churn** model based on their sales, purchased products and browsing behavior



Features

Features are chosen **arbitrarily** (e.g., aggregations, time windows)

Only a **limited set of data** is used

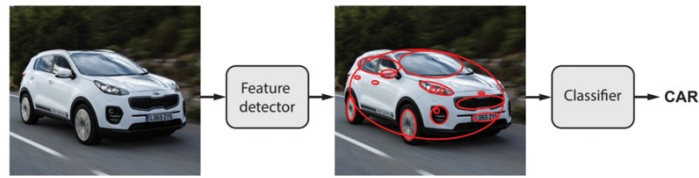
Issues with **point-in-time correctness/information leakage**

Key Question

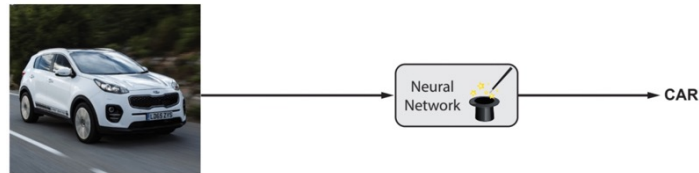
How to leverage the data without going through the **longest duration tasks** (extraction, transformation, loading)?

We want ML algorithms that can process data in its natural form!

CV moved to end-to-end learning



Classical computer vision: hand-crafted features (e.g. SIFT)
+ simple classifier (e.g. SVM)



Modern computer vision: data-driven end-to-end systems

Modern ML

Today we want to design deep learning models that operate on relational tables
But modern deep learning toolbox is designed for
different type of inputs

Doubt thou the stars are fire,
Doubt that the sun doth move,
Doubt truth to be a liar,
But never doubt I love...

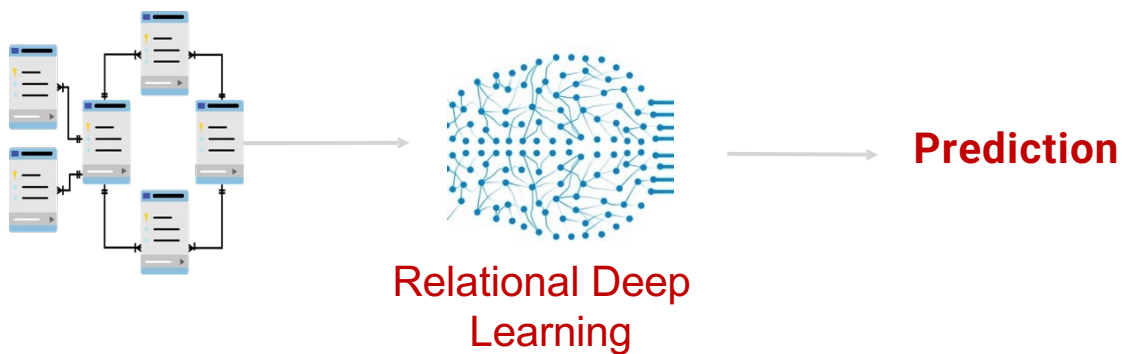
Text



Images



Deep Learning on Relational Tables



End-to-end **deep learning on relational tables** (*i.e.*, databases)

Works directly on relational tables, no transformations, no feature engineering

Casts predictive tasks as **graph representation learning** problems

Graph Neural Networks Meet Relational Databases

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Impact and Consequences

More accurate models

(no feature engineering)

More robust models

(model-learned features automatically “update” over time)

Shorter time to models

(no mundane ETL work)

Simpler infrastructure

(no pipelines, no feature stores, etc.)

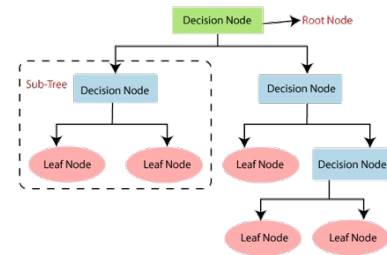
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Related Work: Tabular ML

Great for building models on **one table**

- But most data is not a single table!



Deep learning is not dominant:

- **Hypothesis:** Because single table already contains features engineered from other tables (**loss of information**)

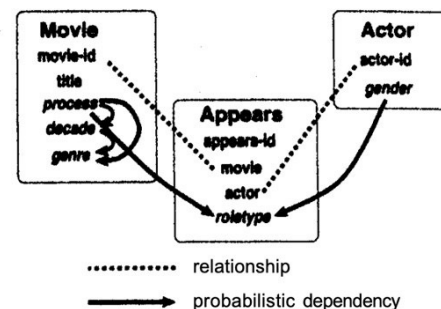
RDL accounts for relations between multiple tables, unlike tabular ML!

Graph Neural Networks Meet Relational Databases

Related Work: Stat. Rel. Learning

[Friedman et al., 1999]

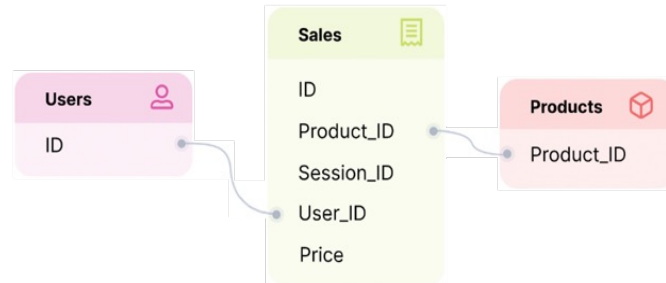
Learns a **distribution** (graphical model) **over the relational structure**



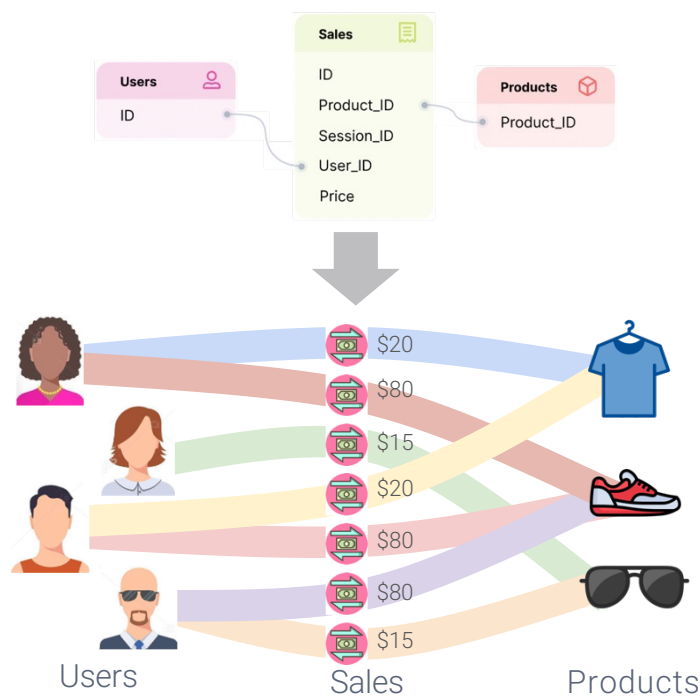
- Predictions via world model + inductive logic programming
- Only works with categorical variables
- Does not scale to large data

RDL a **scalable, expressive inheritor** of Statistical Relational Learning

Insight: A Data is a graph!



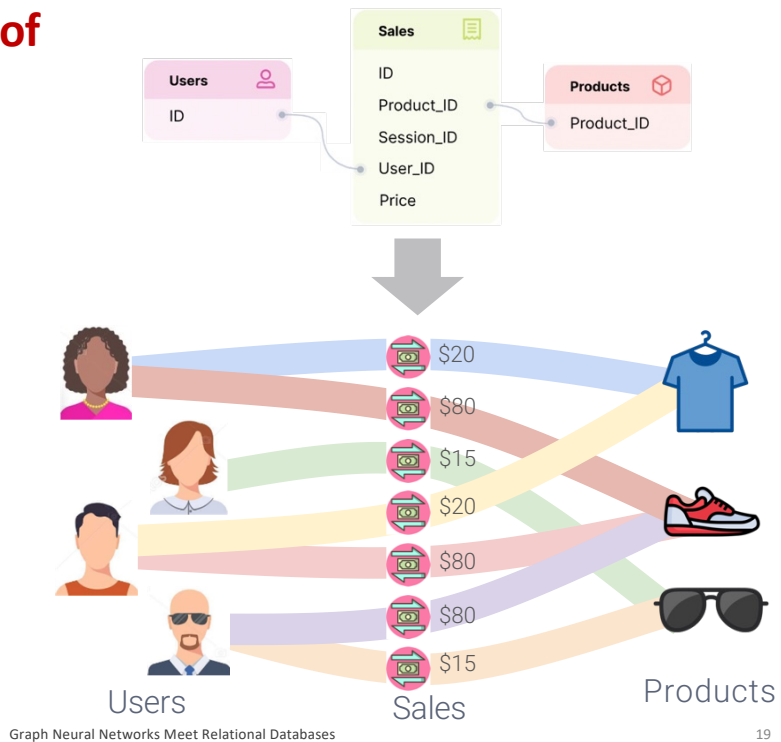
A Database is a graph!



Just do ML on a Graph!

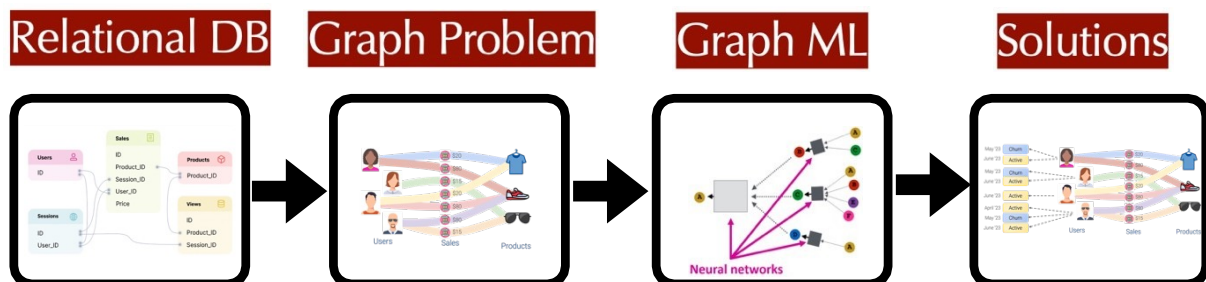
ML in the language of graphs:

- **Node-level:**
 - Churn
 - Life-time value
 - Next best action
- **Link-level:**
 - Product affinity
 - Recommendations
- **Graph-level:**
 - Fraud, money laundering



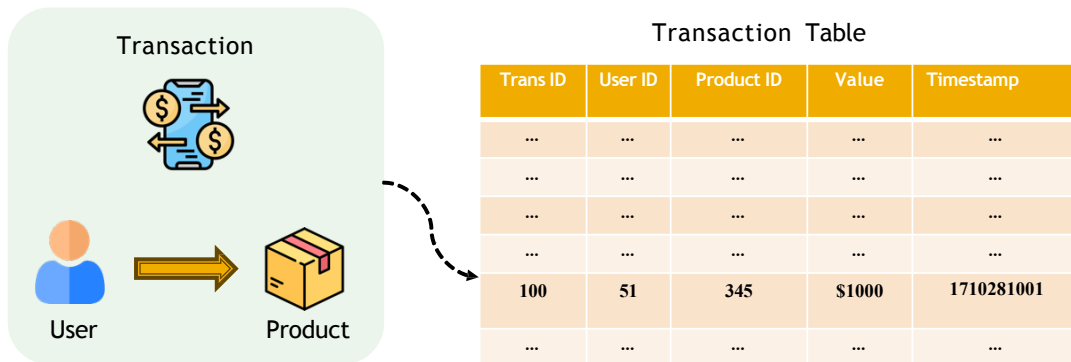
19

Graph ML Problem Solving Pipeline



Databases

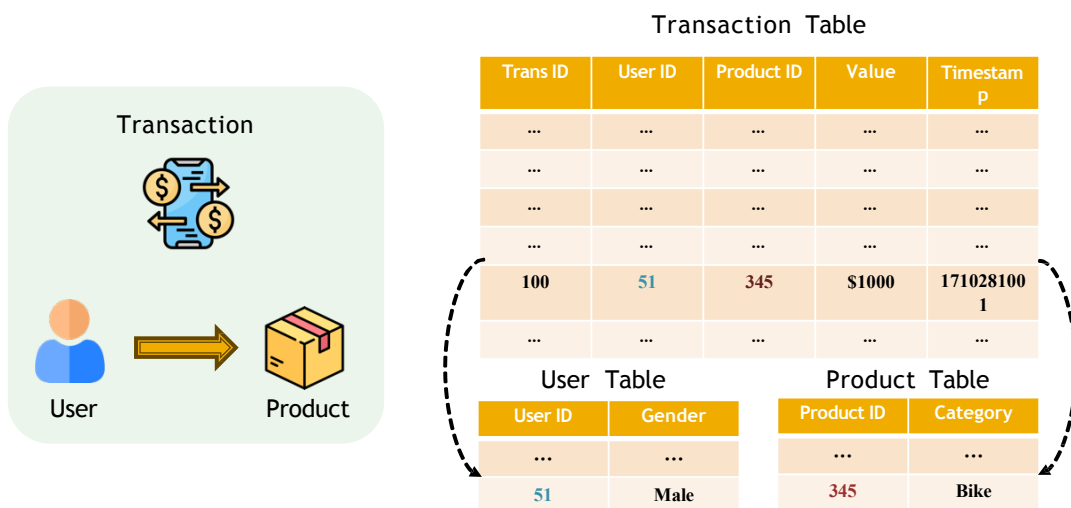
Real-world data are stored in databases



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Relational Databases

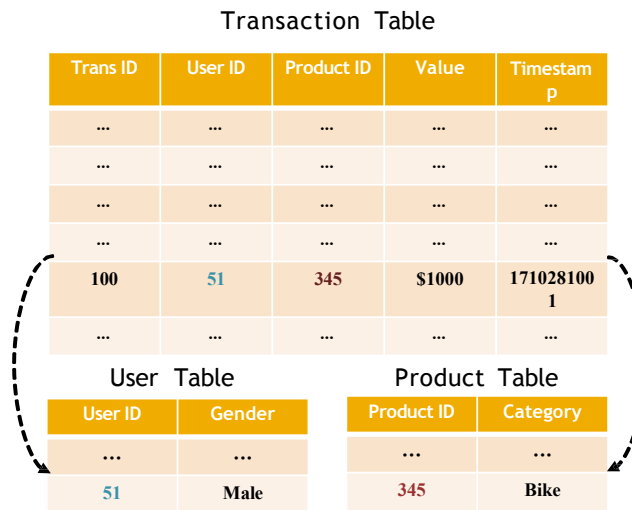
Databases are often relational



Relational databases as heterogeneous graphs

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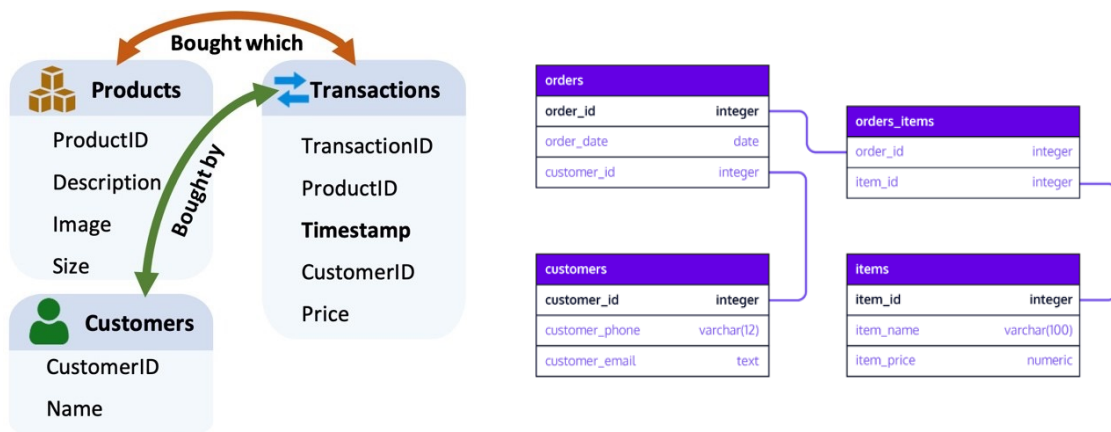
Mathematically...



A database is a set of tables $\mathcal{T} = \{T_1, \dots, T_n\}$ and

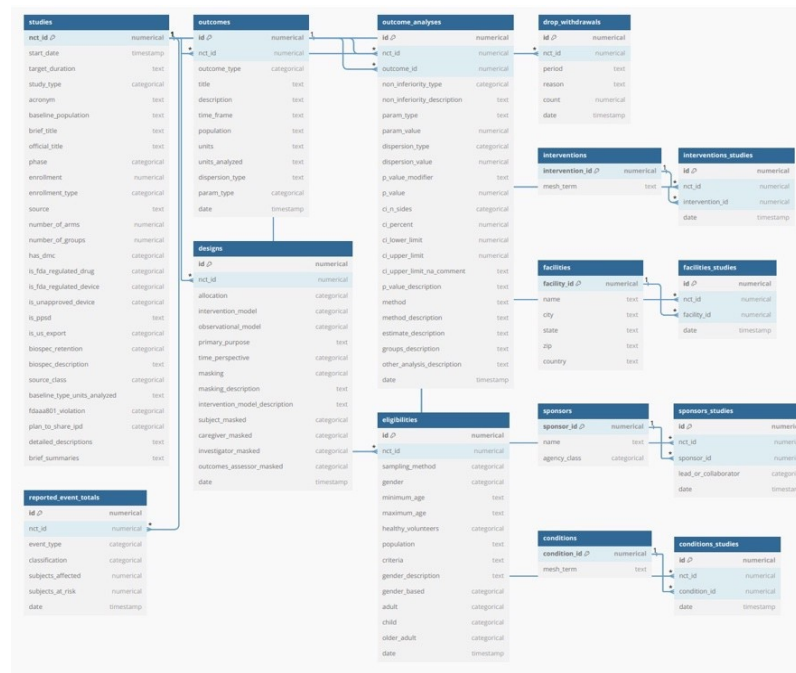
Links between tables $\mathcal{L} \subseteq \mathcal{T} \times \mathcal{T}$

Schema Graph



The schema graph represents the high-level structure of the heterogeneous graph

Rel-trial Schema Graph



The schema graphs can be more complex

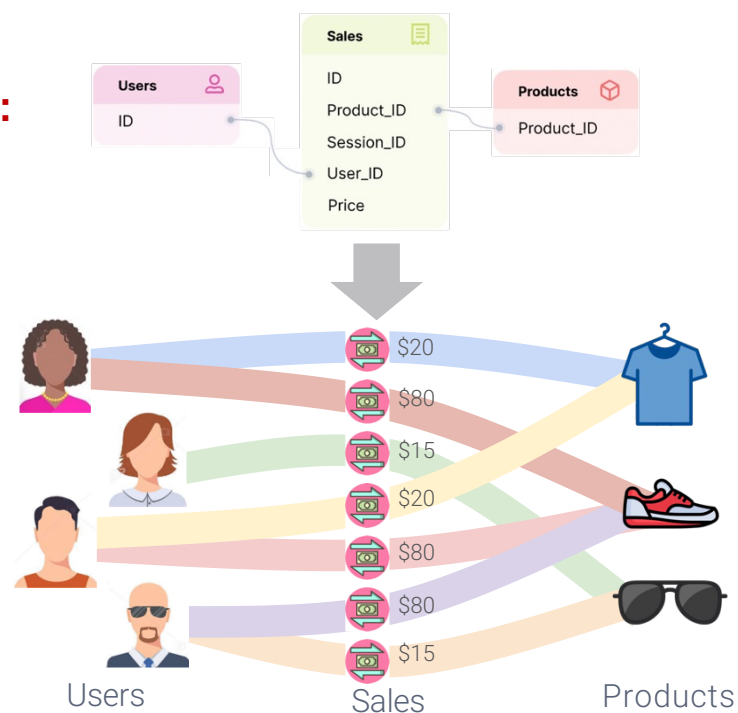
Graph Neural Networks Meet Relational Databases

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Relational Entity Graph

Relational Entity Graph:

Create connections via primary-foreign keys



Graph Neural Networks Meet Relational Databases

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Mathematically...

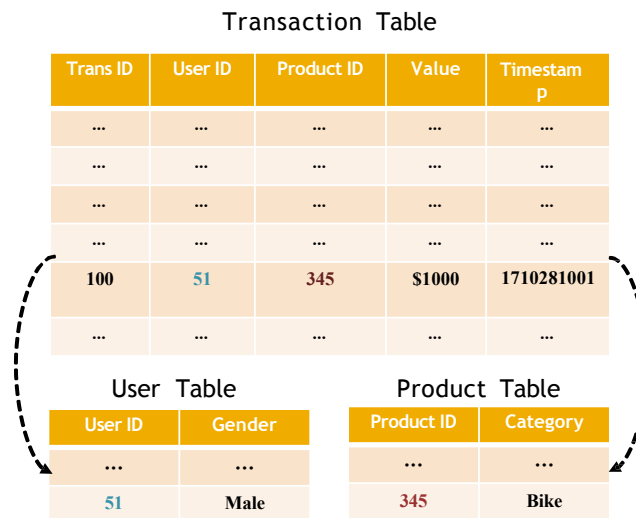
Given database of tables $\mathcal{T} = \{T_1, \dots, T_n\}$ and relations $\mathcal{L} \subseteq \mathcal{T} \times \mathcal{T}$

Each table is a set of entities $v \in T_i$ possessing a primary key and optional foreign keys

The **relational entity graph** is such that

- the **set of nodes** is defined by all rows in all tables
$$\mathcal{V} = \bigcup_{T \in \mathcal{T}} T$$
- The **set of edges** is defined by connecting two entities v_1, v_2 whose primary and foreign keys match

Entities in Relational Entity Graph



Entities also have features (different than KGs)

Overview of the Approach

Predictive Task



Next:

- Define task(s)
- Relational Deep Learning

Defining a Task

- **Example:** Predict whether a user is going to churn in the next 30 days?
- Most tasks are **temporal**:
 - Customer's **label changes over time**
 - Database changes over time
 - For every time, a feature needs to be recomputed
- To define a task, we need:
 - Entity
 - Label
 - Time



Temporal tasks are especially challenging because features are time dependent:

- For every time, a feature needs to be recomputed
- Entity's label can change between time steps

Defining a Task: Training Table

Training Table: A special table containing training labels

- (entity ID, time, labels)
 - Classification, Regression, Multi-class
- time column is essential for **temporal prediction tasks**
 - An entity may have different labels at different times
 - Only use information up to the time of label

Defining a Task: Training Table

Training Table: A special table containing training labels

- (entity ID, time, labels)
 - Classification, Regression, Multi-class

Training Table

Entity ID	Timestamp	Label
99	10172024	1
99	10182024	1
...	...	...
100	10172024	1
100	10182024	0
...	...	...

Example: Churn

Schema:



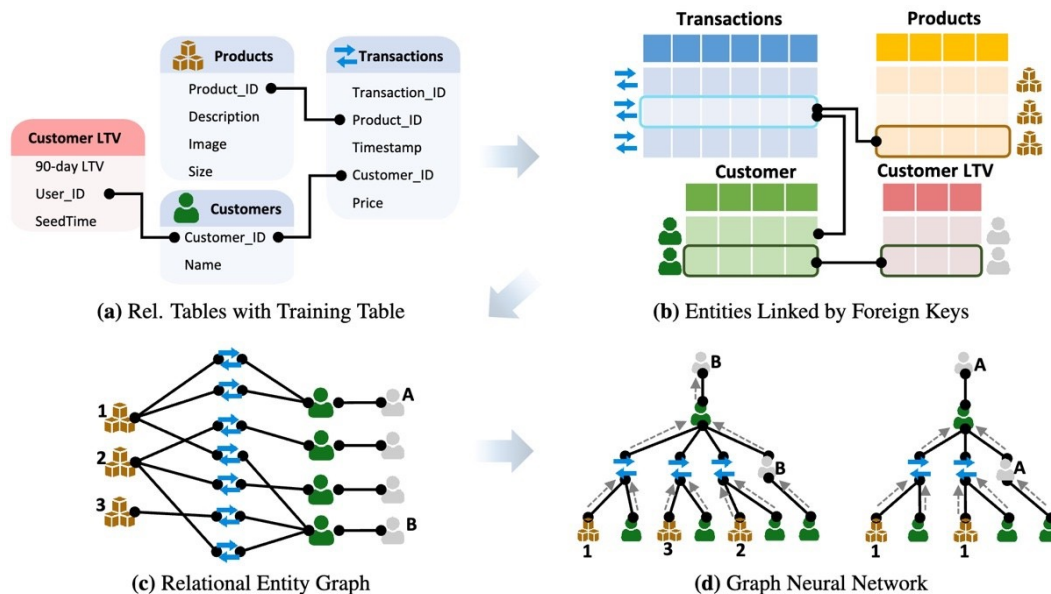
Example prediction task:

- Predict whether a user is going to churn
 - Zero sales in the next 30 days.

Training table:

- (user, time, churn label)

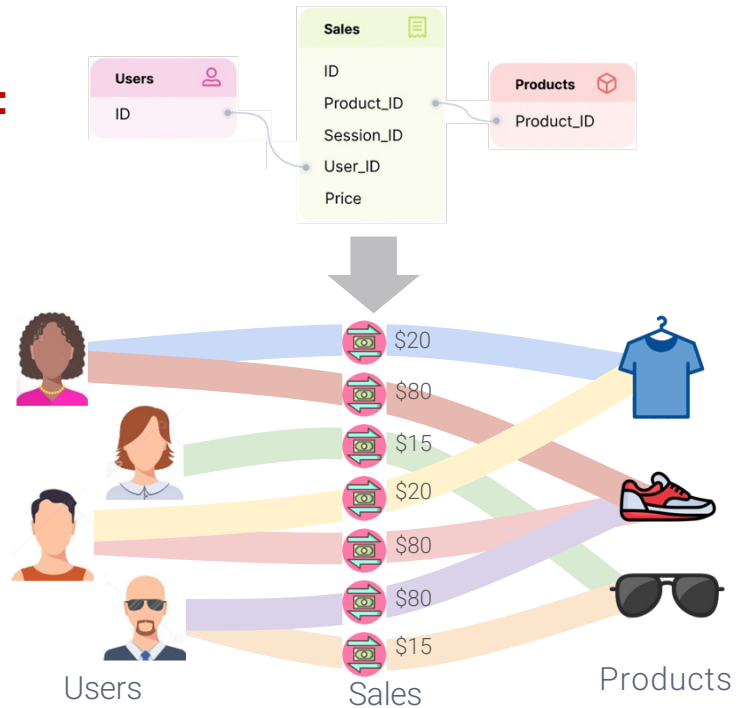
Relational Deep Learning



Relational Entity Graph

Relational Entity Graph:

Create connections via primary-foreign keys

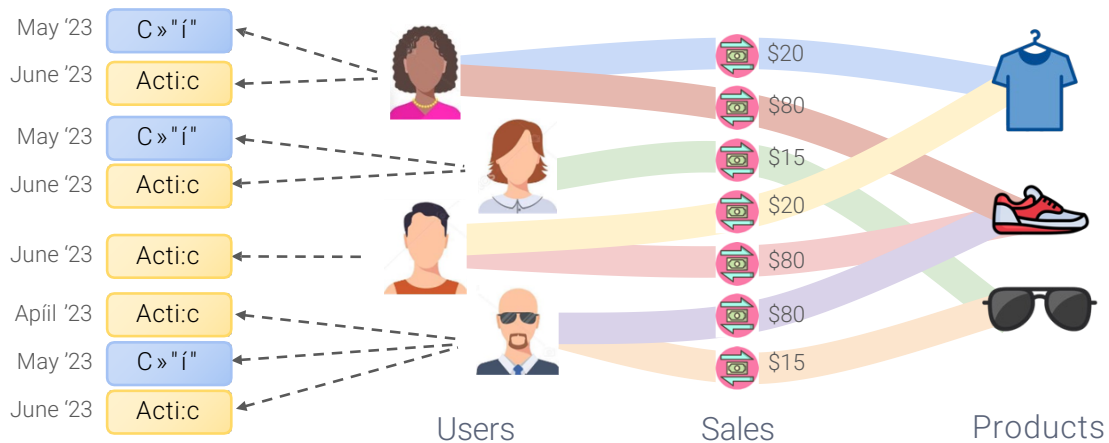


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Connect the Training Table

Training labels together with timestamps are attached to the graph

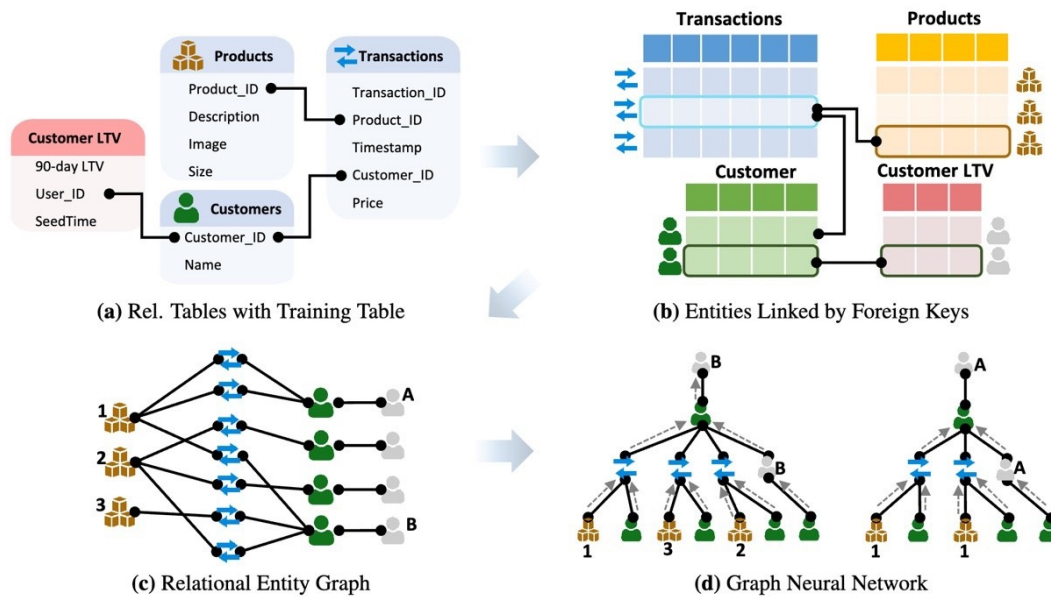


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Relational Deep Learning

Relational GNN



Graph Neural Networks Meet Relational Databases

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GNN on the Entity Graph

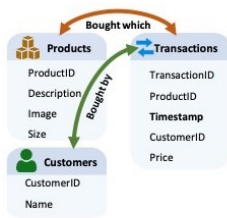
Node's neighborhood defines a computation graph

Nodes learn how to *optimally* use information from neighbors to obtain enhanced node representations

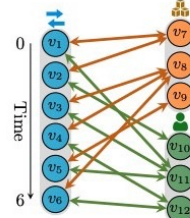


Graph Neural Networks Meet Relational Databases

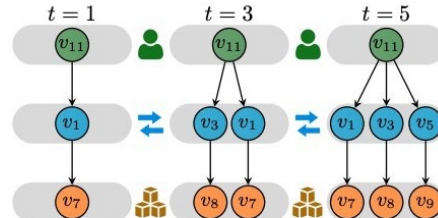
GNNs on Temporal Graphs



(a) Schema Graph



(b) Relational Entity Graph



(c) Computation Graphs for different time t

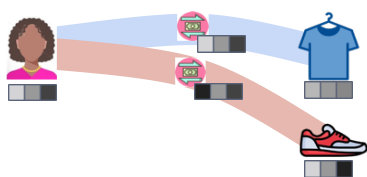
The computation graph for each node is **time-dependent**

Message+Aggregation becomes **time-dependent**

Sampling over neighbors is **time-dependent**

GNN vs Feature Engineering

GNN-based features



VS.

Hand-engineered features

SUM(TRANSACTIONS.Price) o:ci (-30, 0) daQs	AVG(TRANSACTIONS.Price) o:ci (-30, 0) daQs	...
---	---	-----

GNN aggregation is **learnable version of hand-crafted features!**

GNNs give better performance by learning optimal features.

SQL joins vs Graph edges

Definitions:

- A table R is a set of entities $R = \{r_1, \dots, r_n\}$
- Each entity is a tuple $r_i = (r_{i1}, r_{i2}, \dots, r_{in})$
- Given two tables R, S , a join operation is a **subset** of a Cartesian product

- Aggregation will specify which rows are kept

Ex.1: Cross Join

Employee table		Department table	
LastName	DepartmentID	DepartmentID	DepartmentName
Rafferty	31	31	Sales
Jones	33	33	Engineering
Heisenberg	33	34	Clerical
Robinson	34	35	Marketing
Smith	34		
Williams	NULL		

https://en.wikipedia.org/wiki/Relational_algebra

```
SELECT *
FROM employee INNER JOIN department ON 1=1;
```

Employee.LastName	Employee.DepartmentID	Department.DepartmentName	Department.DepartmentID
Rafferty	31	Sales	31
Jones	33	Sales	31
Heisenberg	33	Sales	31
Smith	34	Sales	31
Robinson	34	Sales	31
Williams	NULL	Sales	31
Rafferty	31	Engineering	33
Jones	33	Engineering	33
Heisenberg	33	Engineering	33
Smith	34	Engineering	33
Robinson	34	Engineering	33
Williams	NULL	Engineering	33
Rafferty	31	Clerical	34
Jones	33	Clerical	34
Heisenberg	33	Clerical	34
Smith	34	Clerical	34
Robinson	34	Clerical	34
Williams	NULL	Clerical	34
Rafferty	31	Marketing	35
Jones	33	Marketing	35
Heisenberg	33	Marketing	35
Smith	34	Marketing	35
Robinson	34	Marketing	35
Williams	NULL	Marketing	35

SQL joins vs Graph edges

Connection between SQL Joins and graph edges

Ex.2: Inner Join

$$R \bowtie S := \{r \cup s \mid r \in R \text{ and } s \in S \text{ and } \text{Fun}(r, s)\}$$

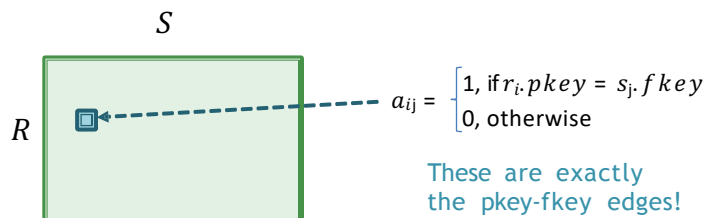
$$\text{Fun}(r, s): r.pkey = s.fkey$$

Employee table		Department table	
LastName	DepartmentID	DepartmentID	DepartmentName
Rafferty	31	31	Sales
Jones	33	33	Engineering
Heisenberg	33	34	Clerical
Robinson	34	35	Marketing
Smith	34		
Williams	NULL		

https://en.wikipedia.org/wiki/Relational_algebra

```
SELECT employee.LastName, employee.DepartmentID, department.DepartmentName
FROM employee
INNER JOIN department ON
employee.DepartmentID = department.DepartmentID;
```

Employee.LastName	Employee.DepartmentID	Department.DepartmentName
Robinson	34	Clerical
Jones	33	Engineering
Smith	34	Clerical
Heisenberg	33	Engineering
Rafferty	31	Sales

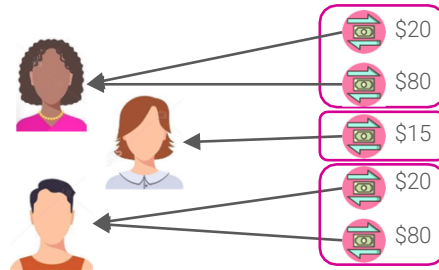


GNNs perform a JOIN+AGG

Input:

USER_ID
1
2
3

USER_ID	VALUE	DATE
1	20	01-01
1	80	01-02
2	15	01-01
3	20	01-02
3	80	01-03



$$\bigoplus_{w \in \mathcal{N}(v)} \mathbf{t}(t_w) \cdot \mathbf{x}_w$$

GNN can learn:

```
SELECT SUM(VALUE)
FROM SALES
WHERE DATE > 01-01
GROUP BY USER_ID
```

USER_ID	SUM(VALUE)
1	80
2	0
3	100

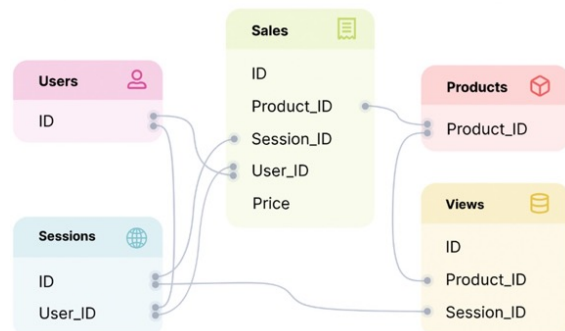
Learnable Aggregation $\bigoplus$
 Temporal embedding $\mathbf{t}(t_w)$
 Fact representation $\mathbf{x}_w$

Benefits of GNNs

GNNs learn how to aggregate information:

They can discard neighboring node information that is irrelevant for the given downstream task

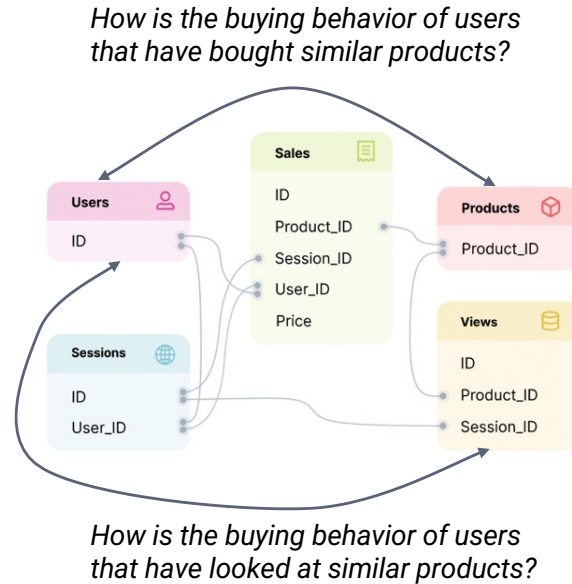
They can detect fine-grained patterns within local neighborhoods (e.g., buying behavior over the last year)



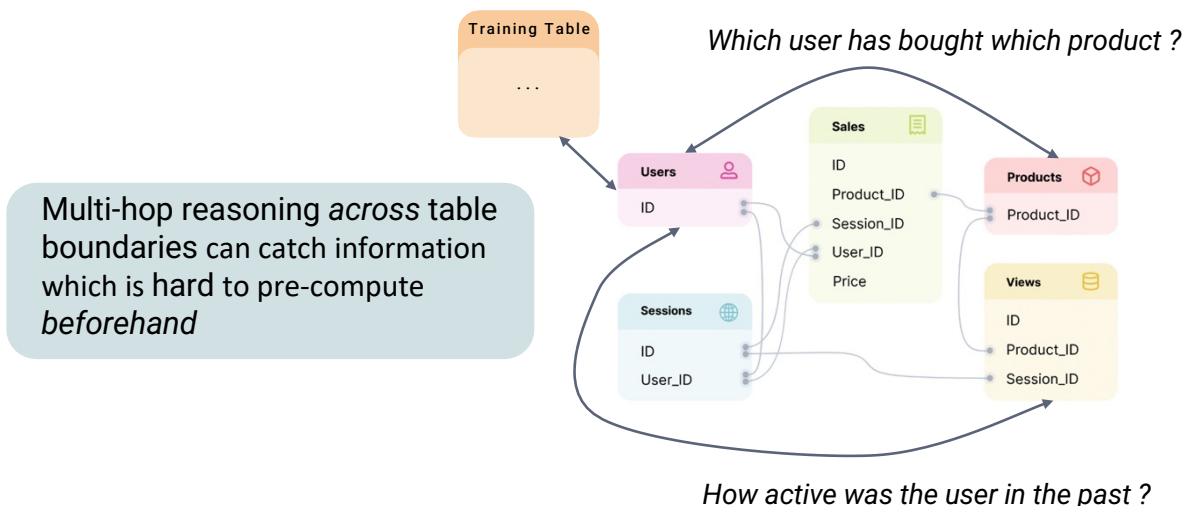
Benefits of GNNs

GNNs can exchange information *across* training examples:

- Instead of treating examples as isolated, there now exists an inter-dependency *between* entities (e.g., users with similar features, users with similar behavior)
- GNN can *use* these features to enrich an entity's representation



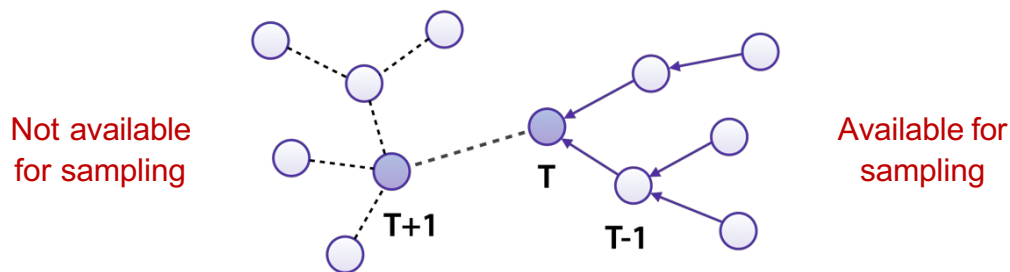
Benefits of GNNs



Benefits of GNNs

First-class temporal support

- Capture fine-grained relative and seasonal features via temporal embeddings
- Avoid data leakage via temporal sampling directly during data loading



Graph Neural Networks Meet Relational Databases

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Full Vision Described in Paper

Relational Deep Learning: Graph Representation Learning on Relational Tables

Matthias Fey^{2,*}, Weihua Hu^{2,*}, Kexin Huang^{1,*}, Jan Eric Lenssen^{2,3,*}, Rishabh Ranjan^{1,*}, Joshua Robinson^{1,*}, Rex Ying⁴, Jiaxuan You⁵, Jure Leskovec^{1,2}

*Equal contribution. Listed in alphabetic order.

¹Stanford University

²Kumo.AI

³Max Planck Institute for Informatics

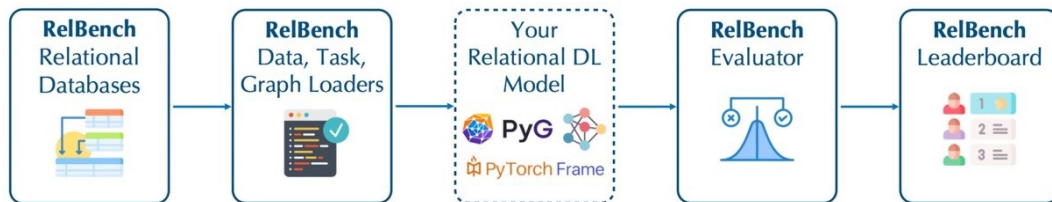
⁴Yale University

⁵University of Illinois at Urbana-Champaign

Available at: <https://relbench.stanford.edu>

Enabling Research on RDL

Relbench is more than just a collection of Databases



Automatically download datasets

Load database and task tables

Standardized evaluation protocol:

- Prevents temporal leakage from test set

Framework-agnostic data structures: use your favorite ML stack!

PyF and PyG Integration

Load as a PyG graph

Train GNN end-to-end

Temporal neighbor sampling

Use PyTorch Frame to encode tables



RelBench Datasets

7 Diverse Datasets



E-Commerce

- rel-amazon
- rel-avito
- rel-hm



Social

- rel-event
- rel-stack



Sports

- rel-f1



Medical

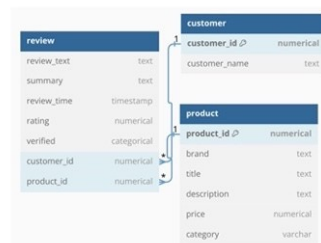
- rel-trial

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RelBench Datasets

Rich Schemas



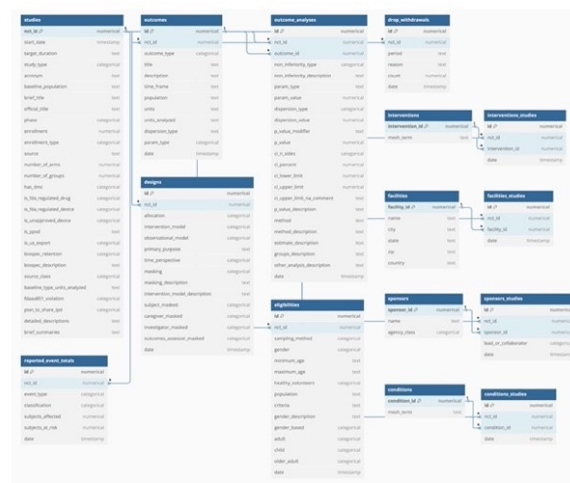
rel-amazon

3 to 15 tables

74k to 41M rows in a DB

15 to 140 columns in a DB

Time span from **2 weeks** to **55 years**



rel-trial

Graph Neural Networks Meet Relational Databases

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RelBench Tasks

30 Real-World Predictive Tasks

Entity Classification

- rel-amazon
 - user-churn
 - item-churn
- rel-stack
 - user-badge
- rel-trial
 - study-outcome
- ...

Entity Regression

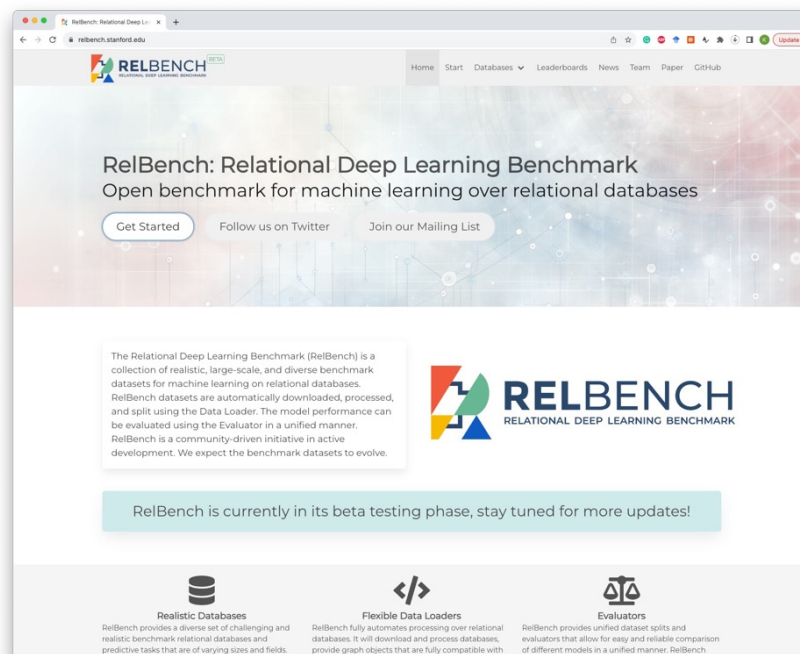
- rel-amazon
 - user-ltv
 - item-ltv
- rel-avito
 - ad-ctr
- rel-f1
 - driver-position
- ...

Recommendation

- rel-amazon
 - user-item-purchase
- rel-avito
 - user-ad-visit
- rel-stack
 - user-post-comment
 - post-post-related
- ...

See website to get started

<https://relbench.stanford.edu>



Expert Data Scientist

GNN versus expert data scientists



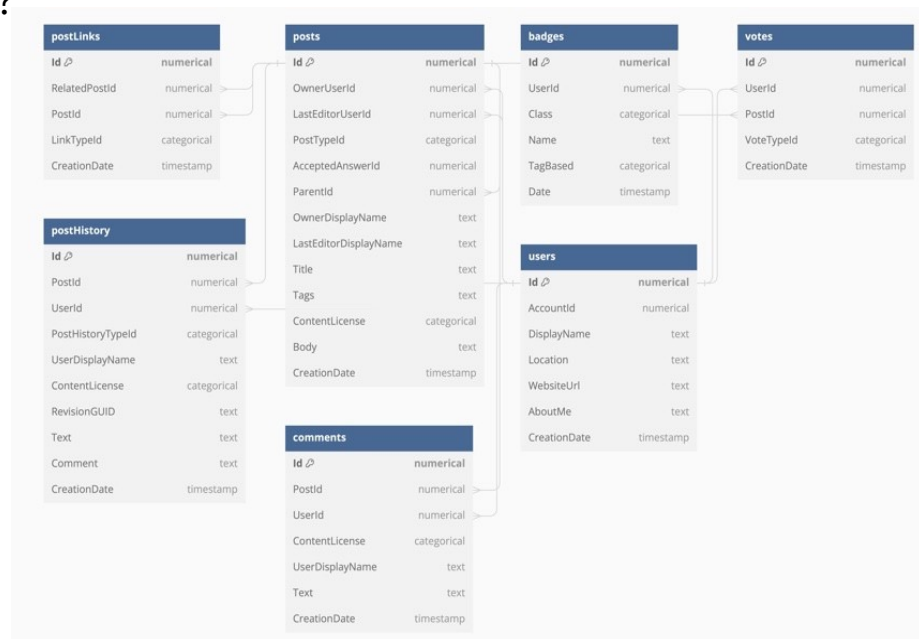
(Alejandro)

- Recruited Experienced Data Scientist
- 5 years in industry (*specializing in financial databases*)
- Responsible for full model building lifecycle (*more detail next*)

Representative Relbench task

Q: Will a user be active in the next 6 months?

Stack Exchange Database



Expert Data Scientist Workflow

Task: Will a user be active in the next 6 months?

Expert Data Scientist Workflow



Exploratory Data Analysis (EDA)

Manual work

4hrs

Task: Will a user be active in the next 6 months?

Expert Data Scientist Workflow



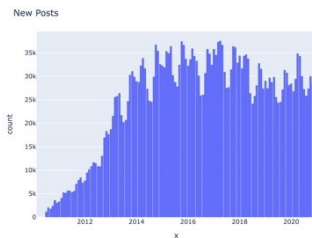
Exploratory Data Analysis (EDA)

Manual work

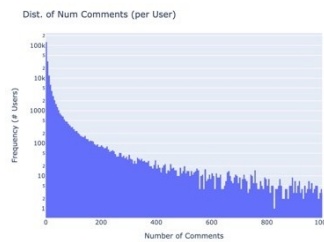
4hrs

Task: Will a user be active in the next 6 months?

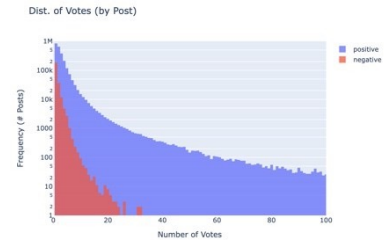
Example observations



Activity is seasonal



Comments follow power law



Negative votes are infrequent

Expert Data Scientist Workflow



Exploratory Data Analysis (EDA)

Manual work

4hrs

Task: Will a user be active in the next 6 months?

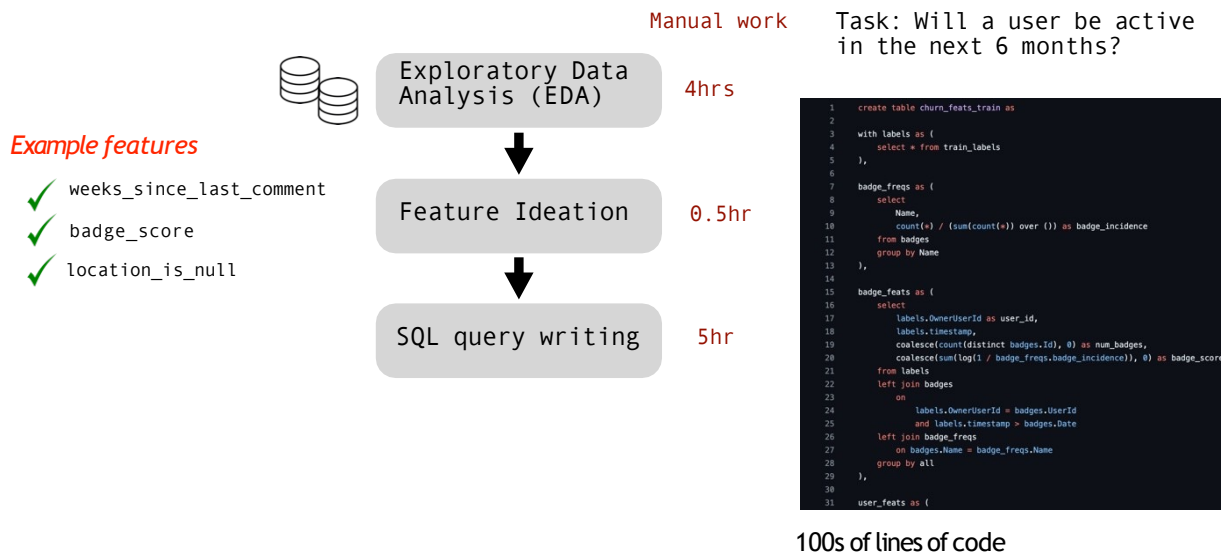
Example features

- ✓ weeks_since_last_comment
- ✓ badge_score
- ✓ location_is_null

Feature Ideation

0.5hr

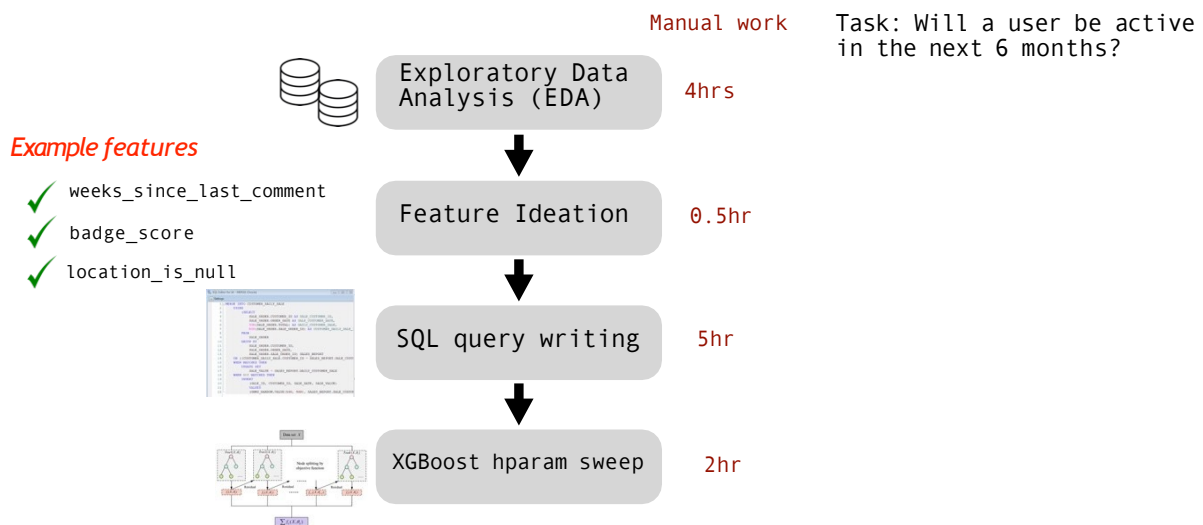
Expert Data Scientist Workflow



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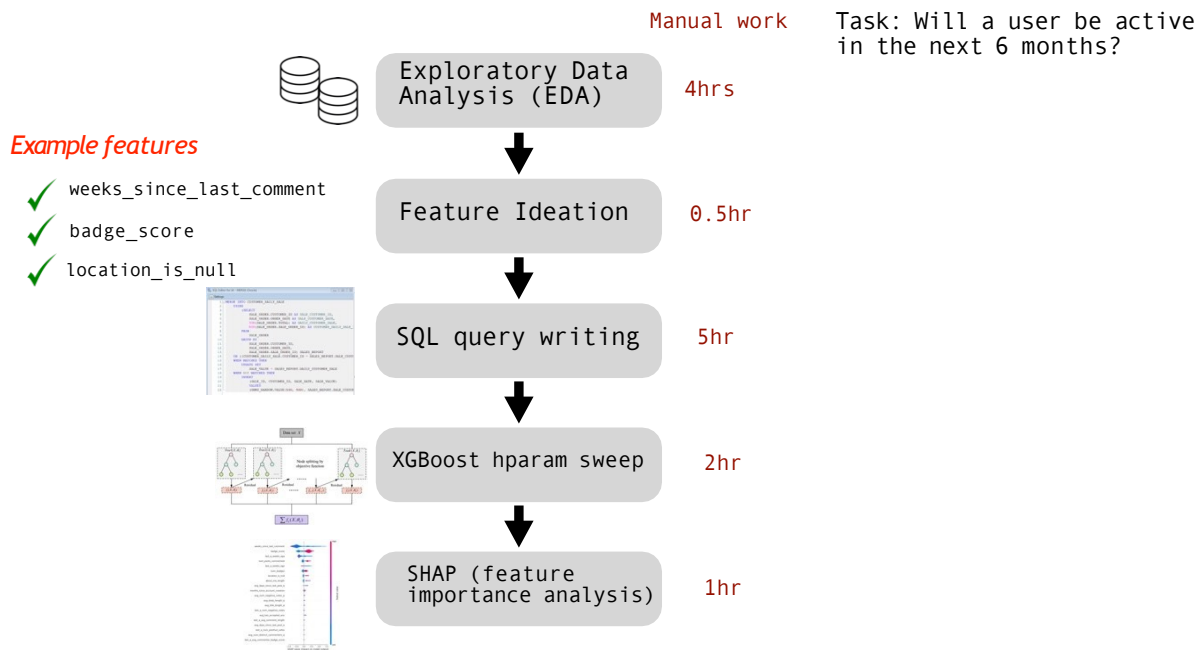
Expert Data Scientist Workflow



Graph Neural Networks Meet Relational Databases

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Expert Data Scientist Workflow



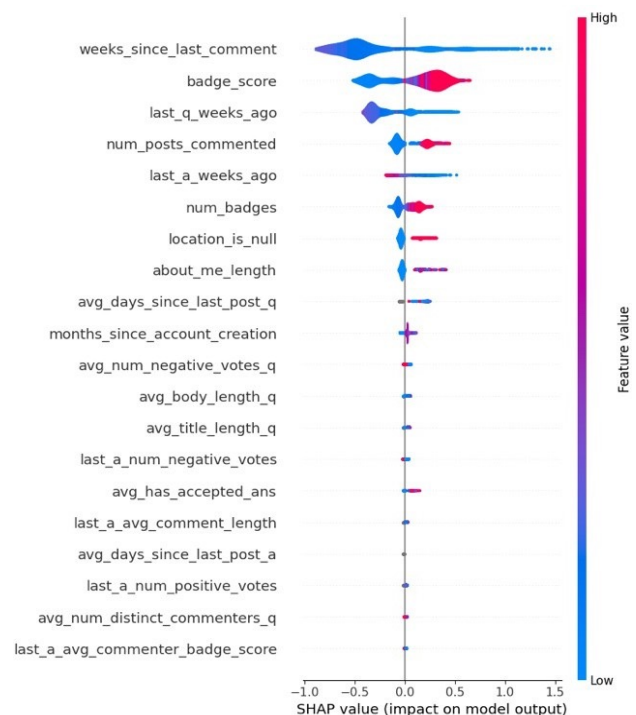
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SHAP feature importance analysis

Selected Observations

- Website “seniority” **predictive** (total number of comments, badge score etc.)
- Time since last active / commented **is predictive**
- Completed bio (about me, location etc.) **is predictive**
- Number of positive/negative votes **not predictive**
- Interacting with “senior” community members **not predictive**



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Final list of features

~12 hours of high-quality expert work

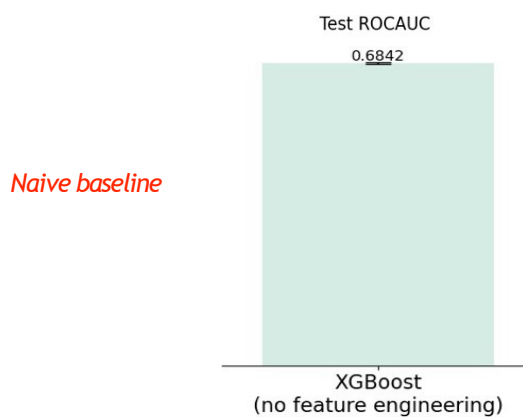
	Label	Class	Label MB	Ratio %
	num_badges	0.229	0.007	0.0%
	num_questions_asked	0.009	0.007	89.7%
	badge_score	0.006	0.009	0.0%
	avg_acceptance_rate	0.195	0.004	96.4%
	avg_comment_length	0.135	0.045	0.0%
	num_positives_comments	0.151	0.041	0.0%
	avg_num_likes	0.131	0.077	89.7%
	avg_has_accepted_ans	0.130	0.106	89.7%
	num_comments	0.130	0.097	0.0%
	num_answers_asked	0.125	0.037	96.4%
	test_u_num_likes	0.111	0.014	13.6%
	avg_num_positives_votes_u	0.108	0.070	96.4%
	test_u_in_accepted_ans	0.108	0.049	73.3%
	test_u_body_length	0.096	0.007	13.6%
	about_me_length	0.096	0.010	0.0%
	avg_num_comments_u	0.084	0.048	96.4%
	test_u_has_accepted_ans	0.076	0.009	13.6%
	avg_num_distinct_comments_u	0.073	0.073	96.4%
	avg_body_length_u	0.067	0.001	89.7%
	avg_num_positives_votes_u	0.060	0.070	89.7%
	test_u_body_length	0.056	0.003	73.3%
	avg_body_length_u	0.034	0.003	96.4%
	test_u_title_length	0.023	0.004	13.6%
	test_u_num_comments	0.023	0.040	73.3%
	test_u_num_comments	0.015	0.003	13.6%
	avg_avg_comment_length_u	0.012	0.047	96.4%
	test_u_avg_comment_length	0.009	0.001	13.6%
	display_name_is_null	0.004	0.001	0.0%
	test_u_num_distinct_comments	0.002	0.067	73.3%
	test_u_num_distinct_comments	-0.003	0.006	13.6%
	test_u_num_positives_votes	-0.006	0.003	13.6%
	test_u_num_positives_votes	-0.009	0.003	73.3%
	test_u_avg_comment_length	-0.010	0.003	73.3%
	avg_num_comments_u	-0.011	0.070	89.7%
	avg_commenter_badge_score_u	-0.015	0.048	89.7%
	avg_title_length_u	-0.020	0.004	89.7%
	test_u_num_negative_votes	-0.026	0.002	13.6%
	avg_num_distinct_comments_u	-0.037	0.046	89.7%
	avg_avg_comment_length_u	-0.043	0.007	89.7%
	test_u_avg_commenter_badge_score	-0.046	0.006	13.6%
	test_u_avg_commenter_badge_score	-0.048	0.003	73.3%
	avg_commenter_badge_score_u	-0.060	0.003	96.4%
	website_url_is_null	-0.061	0.002	0.0%
	test_u_num_negative_votes	-0.065	0.001	73.3%
	months_since_account_creation	-0.113	0.011	0.0%
	avg_num_negative_votes_u	-0.114	0.100	89.7%
	location_is_null	-0.136	0.001	0.0%
	avg_days_since_test_post_u	-0.168	0.005	97.1%
	avg_days_since_test_post_u	-0.169	0.005	95.5%
	avg_num_negative_votes_u	-0.170	0.088	96.4%
	test_u_weeks_ago	-0.340	0.073	13.6%
	weeks_since_test_comment	-0.360	0.001	34.3%
	test_u_weeks_ago	-0.364	0.045	73.3%
	test_u_num_likes	nan	0.065	73.3%

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Results

Task: Will a user be active in the next 6 months?

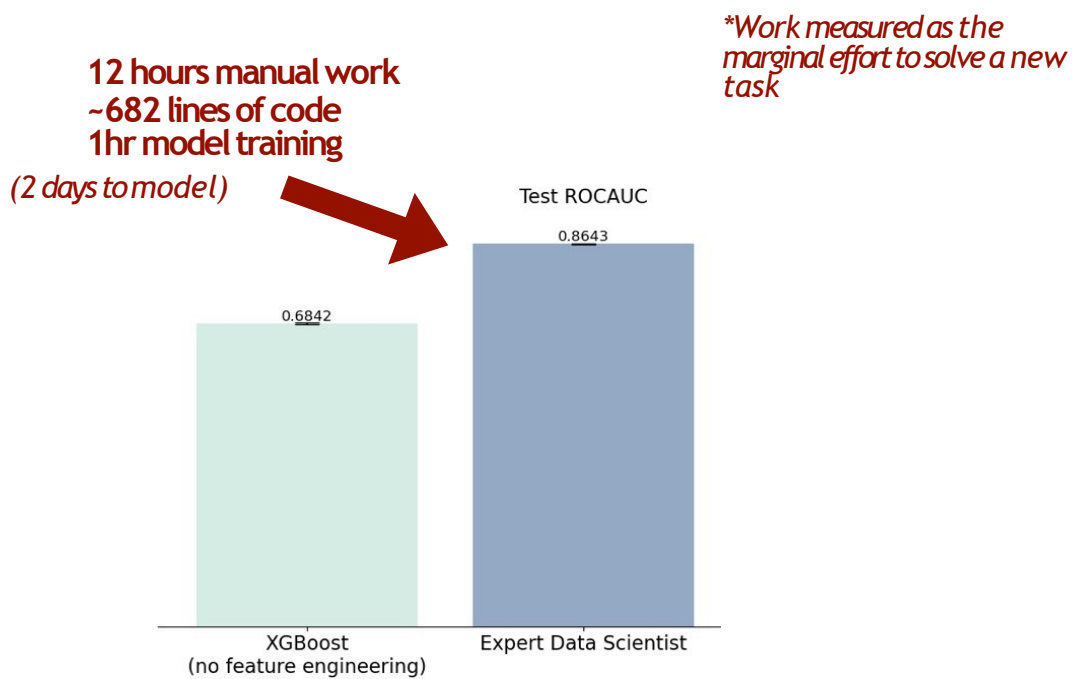


users	
Id	numerical
AccountId	numerical
DisplayName	text
Location	text
WebsiteUrl	text
AboutMe	text
CreationDate	timestamp

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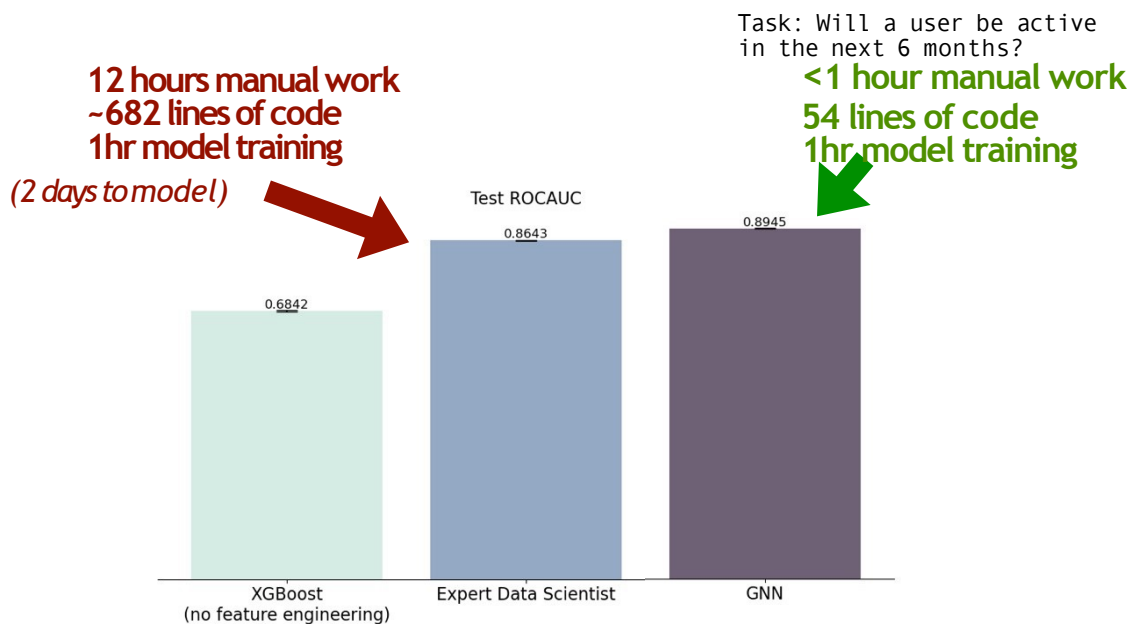
Results



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Results

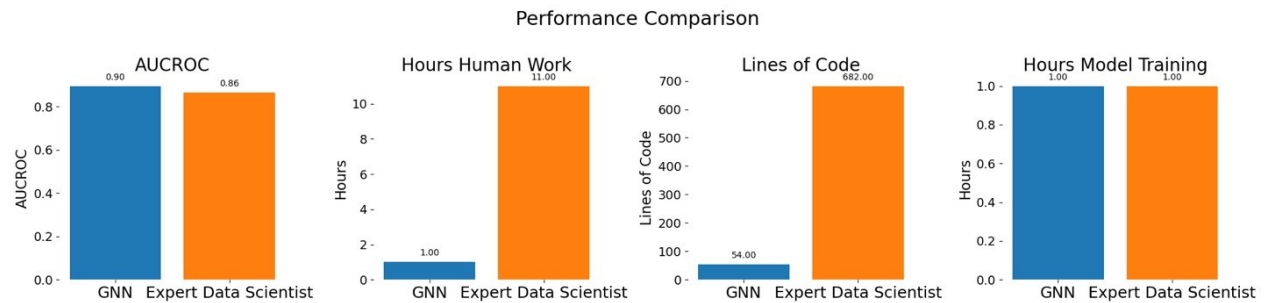


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Results

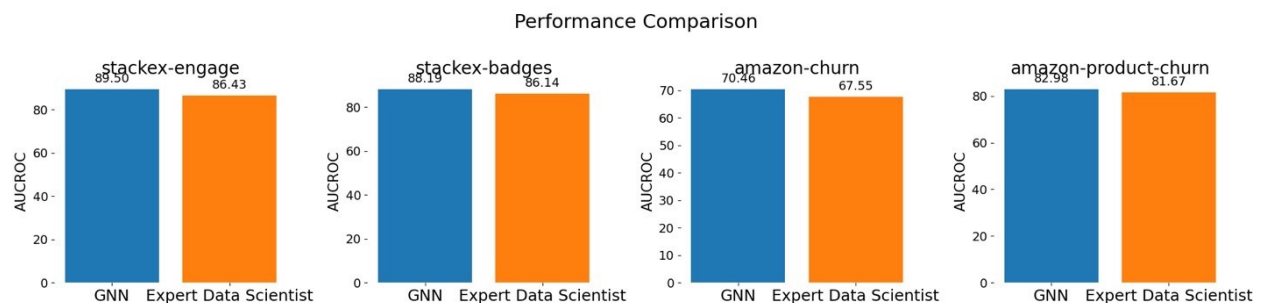
Task: Will a user be active in the next 6 months?



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More Relbench tasks



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